Servio: Restaurant POS Product Spec and Build Blueprint

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Summary

Servio is a cloud restaurant POS sold as a monthly subscription in Pakistan and internationally. Each restaurant signs up, gets a 3-month free trial, and sees only its own data. It matches ArfaPOS on features, but is simpler to use, safer with cash, and tax-ready from day one.

This document is the build spec for the development team. Part 1 reviews ArfaPOS (bizpos.airoxlab.com), the product Servio replaces, based on 21 admin screenshots from 7 Oct 2026. Part 2 covers brand, pricing, modules, business logic, accounts and data isolation, WhatsApp, data model, roles, design, tech stack, security, process and roadmap. Where this document is silent, developers should ask, not guess.

Our three edges over ArfaPOS:

- **Simple by default.** A small cafe sees 6 menu items, not 25. Advanced modules switch on when needed.
- **Cash and fraud control.** No order deletes, only voids with reason. Shift-based cash drawer with day-end report.
- **Pakistan-ready.** FBR and PRA invoice integration, Foodpanda orders, EasyPaisa, JazzCash, Urdu receipts.

Part 1: ArfaPOS overview

ArfaPOS is a web admin panel with 20 top-level sections, plus a till app, tablet app, customer website and app. The inventory side is its strongest part. Sections marked "not seen" exist in the sidebar but were not opened.

Module	What it contains	What it does	Seen
Dashboard	Products, revenue, open orders, customers, expenses; 6-month sales trend; weekly orders; category and status charts; low-stock alert	One-screen health check of the business	Yes
Products	Menus, Catalog, Deals, Promotions, Quick Add Menu	Multiple menus (e.g. Ramadan Iftar) with show/hide online; 134 products in categories; combo deals with images; limited-time promotions	Yes
Orders	Order cards with serial (#59), type, customer, rider, payment method, status; filters; Mark All Completed; Export CSV	Live list of today's orders across walk-in, takeaway, delivery	Yes
Expenses	Categories, subcategories, payment method, by cashier or admin; PDF and CSV export	Records daily spending	Yes (empty)
Inventory	Stock in Hand, Stock Actions, Demand, Items, Recipes, Order Inventory, Units, Inventory Settings	Stock by section (Kitchen, Warehouse); receive, transfer, waste, count; reorder suggestions; recipes with yield and cost; packaging per order type; unit conversions	Yes
Inventory Reports	Daily Stock Sheet, Stock Ledger, Consumption, Waste and Adjustments, Actual vs Theoretical, Valuation and COGS, AI Stock Forecast, Deduction Health	Full audit of every stock movement with who punched and who completed; flags recipes with odd amounts	Yes
Payroll	Overview, Employees, Advances, Run Payroll, Ledger, Fingerprint, Reports, Settings	Monthly payslips with advances, bonuses, tips; fingerprint attendance	Yes

Module	What it contains	What it does	Seen
Purchasing		Purchase orders and goods received (expected)	Not seen
Suppliers		Supplier list and balances (expected)	Not seen
Customers		881 saved customers	Not seen
Staff		Users and roles (expected)	Not seen
Finance		Cash, bank, accounts (expected)	Not seen
Reports		Sales reports (expected)	Not seen
Riders		Delivery riders and assignment	Not seen
Tables		Dine-in table layout	Not seen
QR Ordering		Customers order by scanning a table QR	Not seen

Key settings found in Inventory Settings:

- **Day lock.** Stock changes before a locked date are refused, but sales still go through.
- **Counting switch.** Stock counts only open during a stocktake.
- **Below-zero rule.** Manual moves cannot go below zero; sales and waste always can.
- **Return on cancel.** A cancelled order puts its ingredients back.
- **Safety net.** Orders whose stock deduction failed (offline till, slow internet) are found and deducted every few minutes.

Strengths and weaknesses

ArfaPOS is feature-rich but hard for a normal owner, and its live data shows the setup is not working. We copy the strengths and design out the weaknesses.

Strengths to copy

- Stock Ledger records every movement with order number, "punched by" and "completed by". Great for catching staff issues.

- Consumption report flags ingredients using more than 10x the normal amount per order. Catches recipe mistakes.
- Day lock, return-on-cancel and safety-net settings are well thought out.
- Multiple menus with online on/off switch suit Ramadan and seasonal menus.
- Local payments (Cash, EasyPaisa, Alfalah, Account) and PKR everywhere.
- Clear help text on every page.

Weaknesses to fix in our product

Problem seen	Why it matters	Our fix
40 items below zero, Rs 153,467 short	All inventory reports become wrong	Guided opening-stock setup; alert when an item goes negative; daily "negative stock" task for the manager
"Labor Cost" and "Gas Cylinder" kept as stock items	Food cost looks wrong; labor is not inventory	Separate Overheads setting (labor, gas, rent) applied as cost per order or per month
Trash icon on every order	Deleting orders is the most common cash theft	No delete. Void only, with reason and manager PIN, shown in a Void report
"Mark All Completed" button	Completes unpaid orders and riderless deliveries in one click	Removed. Bulk actions only on paid orders, with confirmation
2 recipes for 134 products	Stock is not deducted for most sales	Recipe coverage meter on dashboard; product cannot go live without a recipe (optional rule)
Product count 162 vs 134 vs 161	Owner cannot trust numbers	Clear names: "Menu items" and "Stock items"
Weekly Orders chart axis shows "Rs"	Basic label bug	Design review checklist for every chart
0.0833333333 in conversions	Looks unprofessional	Show 3 decimals or fractions (1/12)

Problem seen	Why it matters	Our fix
Conversions saved both ways	Two rules can disagree	Save one rule, calculate the reverse
20+ sidebar sections	New users feel lost	Role-based menu; modules switched on in Settings
Content cut off on 1366px screens	Buttons hidden	Test every page at 1366px, 1024px tablet, 390px phone
Expenses at Rs 0	Profit reports meaningless	Daily expense prompt at shift close
Three names: bizpos, ArfaPOS, airoxlab	Weak brand	One name, one domain

Part 2: Our product vision

A cashier should take an order in under 10 seconds, and an owner should understand the day in under 1 minute.

Target customers

- Small cafes, tea shops and fast food (1 counter, 2 to 5 staff)
- Desi restaurants with dine-in, takeaway and delivery (10 to 40 staff)
- Chains with 2 to 20 branches and a central kitchen

Design principles

1. **Simple first, deep later.** Start with Orders, Menu, Cash and Reports. Inventory, Payroll and Multi-branch are switches in Settings.
2. **Every rupee has a trail.** Every order, void, discount, refund and cash drawer action is logged with user, time and reason. Nothing is deleted.
3. **The till never stops.** Works offline, syncs later. A sale is never blocked by a stock figure or slow internet.
4. **Numbers you can trust.** One name per number, the same value on every screen.
5. **Pakistan first.** PKR, Urdu receipts, FBR/PRA, Foodpanda, EasyPaisa, JazzCash, load-shedding tolerant.
6. **Fast to set up.** Import menu from Excel or the old POS export (photo import with AI later). Live in one day.

Brand

The product name is **Servio**, with one global English tagline and local versions per market.

Use	Tagline
Global master (website, app stores)	Servio. Every order. Every payment. Under control.
Short (logo, app icon, social bio)	Serve faster. Earn smarter.
Pakistan, English	Every order. Every rupee. Under control.
Pakistan, Urdu (ads, WhatsApp)	Har order, har rupee ka hisaab.
Gulf, Arabic (later; check with a native speaker)	كل طلب. كل ريال. تحت السيطرة.

Domains

- Main: servio.com, or getservio.com / servio.app if taken.
- Pakistan: servio.pk, pointing to a Pakistan page with PKR prices and the Urdu tagline.
- Social handles: @servio or @getservio on Instagram, Facebook, LinkedIn, TikTok.
- Before launch: trademark search on IPO Pakistan, then register the name.

Pricing and free trial

Three plans, priced per branch per month at the low end of the market to win customers from costlier POS systems. Every new account gets 3 months of Pro free. Plans show the full feature set; features arrive by phase (see Build roadmap), so early customers keep their launch price as new features are added.

Plan	Pakistan	International	Included	WhatsApp utility messages / month
Starter	PKR 2,500	$19	1 till, menus and deals, orders, payments, cash drawer, basic reports, up to 3 staff	300

Plan	Pakistan	International	Included	WhatsApp utility messages / month
Pro (most popular)	PKR 4,999	$39	Starter + kitchen display, tables, riders, inventory and recipes, FBR/PRA tax, online and QR ordering, unlimited staff	1,000
Business	PKR 8,999	$69	Pro + multi-branch dashboard, payroll, loyalty, Foodpanda orders, own WhatsApp number, custom domain, priority support	3,000

Extras

- Extra till or tablet: PKR 1,000 / $9 per month.
- Yearly billing: pay 10 months, get 12.
- Optional setup and menu entry: PKR 5,000 / $49 one-time.
- WhatsApp marketing messages: prepaid credits, priced at Meta's rate plus a small margin.

Trial rules

1. Signup gives full Pro features for 90 days, no card needed.
2. Reminders by WhatsApp and email on day 60, 80 and 88.
3. Paying yearly before the trial ends adds 1 extra free month.
4. After 90 days without payment, the account becomes read-only: data can be viewed and exported, but no new orders. Data is kept 60 more days, then deleted after a final warning.
5. One trial per phone number, email and business name.

Billing

- Plans, add-ons and limits live in the database, not in code, so prices can change without a release.
- Pakistan: JazzCash, EasyPaisa, bank transfer; monthly invoice with manual or automatic confirmation.
- International: Paddle as merchant of record (cards, global tax, payouts to Pakistan).

- Failed payment: 7-day grace with reminders, then read-only.
- Upgrade takes effect at once (prorated); downgrade at the next billing date.

Modules

Servio has 22 restaurant modules in 6 groups, plus 6 platform modules listed after the table. Modules marked "New" are missing in ArfaPOS; "Core" ones ship in the first version.

Group	Module	Key logic	vs ArfaPOS	Phase
Selling	Till (cashier screen)	Category grid, search, modifiers (size, spice, extra cheese), split bill, hold order, discounts with limits per role, works offline	Same, improved	Core
Selling	Kitchen Display and KOT	Orders routed by station (BBQ, Karahi, Tandoor, Drinks); timer per order; bump when ready; KOT printing fallback	New	Core
Selling	Tables and Dine-in	Floor plan, table status colors, merge and move tables, waiter assignment, service charge	Same	Core
Selling	Delivery and Riders	Rider assignment, cash collected per rider, rider settlement at shift end, delivery zones and charges	Same, improved	Phase 2
Selling	Online ordering, QR menu, app	Website and QR ordering from same menu; online on/off per menu, category, item	Same	Phase 2
Selling	Aggregator orders	Foodpanda and others pulled into the same order list, with commission tracking	New	Phase 3
Selling	Reservations	Table booking, deposit, reminder SMS/WhatsApp	New	Phase 3

Group	Module	Key logic	vs ArfaPOS	Phase
Menu	Menus and Catalog	Multiple menus, categories, items, variants, modifiers, images, time-based menus (breakfast, Sehri)	Same	Core
Menu	Deals and Promotions	Combo deals, happy hours, % or fixed discount, coupon codes, date and time rules	Same	Core
Money	Shifts and Cash Drawer	Opening cash, cash in/out with reason, blind closing count, over/short, Z report per cashier	New	Core
Money	Tax (FBR / PRA / SRB)	Tax per item or category; real-time invoice sync; tax invoice number and QR on receipt; offline queue	New	Core
Money	Payments	Cash, card, EasyPaisa, JazzCash, bank transfer, customer credit (khata), split payment	Same, improved	Core
Money	Expenses	Categories, receipts photo, petty cash from drawer, daily prompt at shift close	Same, improved	Phase 2
Money	Finance and Accounts	Cash and bank accounts, supplier payables, customer receivables, profit and loss	Same	Phase 3
Stock	Inventory	Stock items, sections/stores, receive, transfer, waste, count, day lock, below-zero alerts	Same, improved	Phase 2
Stock	Recipes and Costing	Recipe per menu item and sub-recipe (sauce, dough), yield, auto food cost %, recipe coverage meter	Same, improved	Phase 2
Stock	Purchasing and Suppliers	Purchase orders, goods received, supplier price history, reorder suggestions	Same	Phase 2

Group	Module	Key logic	vs ArfaPOS	Phase
People	Customers and Loyalty	Customer profile from phone number and order history (Core); points, birthday offers, WhatsApp campaigns (Phase 3)	New (loyalty)	Phase 3
People	Staff, Roles, Attendance	Users, PIN login, roles (Core); fingerprint or selfie attendance and shift schedule (Phase 3)	Same, improved	Core
People	Payroll	Salary, advances, bonuses, tips share, late/absent deductions from attendance	Same	Phase 3
Insight	Reports and Dashboard	Sales, items, hours, cashiers, voids, discounts (Core); menu profitability (Phase 2); branch comparison (Phase 3)	Improved	Core
Insight	Multi-branch and Central Kitchen	Branch setup, central menu with branch prices, stock transfers, owner view of all branches	New	Phase 3

Platform modules (SaaS layer)

These 6 modules turn the POS into a product many restaurants can buy. All ship in Phase 1, because nobody can use Servio without them.

Module	Key logic	Phase
Signup and onboarding	Sign up with phone (WhatsApp OTP) or email; business name, country, currency, time zone; pick sample menu or import; setup checklist with progress bar	Core
Accounts and branches	One business account holds many branches; each branch has its own tills, tables, stock sections and tax settings	Core

Module	Key logic	Phase
Users and login	One login can belong to several businesses with a different role in each; staff use PIN on registered devices	Core
Subscription and billing	Plans, trial, invoices, payments, grace period, read-only mode, feature limits per plan	Core
WhatsApp and notifications	Receipts, order status, owner daily report, OTP, reminders, opt-in marketing	Core (utility), Phase 3 (marketing, bot)
Super admin (Servio team)	List of all businesses, plans, trial status, revenue, support login with customer approval, feature flags, system health	Core

Additional features

These features complete the modules above. Many are everyday needs in Pakistani restaurants.

Feature	What it does	Module	Phase
Settings module	Business info, branches, business hours, day-end time, tax setup, payment methods, printers and stations, receipt designer, languages	Platform	Core
Receipt designer	Logo, header, footer, social links, English or Urdu, 58/80 mm preview	Settings	Core
Token / order-ready screen	Customer-facing screen showing "Order #59 ready"; sound alert	Till, Kitchen	Core
Prices by order type	Different price for the same item in dine-in, takeaway and delivery	Menu	Core
Staff meals and complimentary orders	Free orders with reason and approver; cost tracked, not hidden	Till	Core

Feature	What it does	Module	Phase
Kitchen change tickets	Edit or cancel after sending prints "CHANGED" or "CANCELLED" ticket to the station	Kitchen	Core
Open item	Sell an item not on the menu with a custom name and price (permission needed)	Till	Core
Price override	Change an item's price on the spot, manager PIN only, logged	Till	Core
Cash rounding	Round bill to nearest Rs 5 or Rs 10; rounding shown on receipt	Payments	Core
Raast QR payment	Raast QR on screen or receipt; customer pays from any bank app	Payments	Core
Card terminal link	Amount sent to the bank card machine so it is not typed by hand	Payments	Core
Reprint receipt	Reprints marked "DUPLICATE", logged	Till	Core
Tips	Tip added on bill, shared to staff in payroll	Payments	Core
Customer queue display	Second screen showing current order and total	Till	Phase 2
Phone orders with caller ID	Incoming call shows customer name, address and last order	Delivery	Phase 2
Bulk and catering orders	Pre-orders for a future date (e.g. 10 degs of biryani) with advance payment	Orders	Phase 2
Advance payments and deposits	On orders and reservations; adjusted in the final bill	Payments	Phase 2
Course firing	Send starters now, mains later, for dine-in	Kitchen	Phase 2
Combos with choices	"Burger deal: choose any drink and any side"	Menu	Phase 2

Feature	What it does	Module	Phase
Items sold by weight	BBQ, sweets, bakery by kg with weighing scale	Till, Stock	Phase 2
Batch and expiry dates	For meat, dairy, sauces, with expiry alerts	Stock	Phase 2
Stock count on mobile	Count stock with a phone while walking the store	Stock	Phase 2
Supplier bill photo scan	Photo of supplier bill fills the goods received form (AI)	Purchasing	Phase 2
Waiter sales and commission	Sales per waiter, commission rules	Staff	Phase 2
Scheduled reports	Daily, weekly, monthly reports sent by WhatsApp or email	Reports	Phase 2
Tax filing report	Monthly sales tax summary for the accountant	Reports	Phase 2
Cloud kitchen brands	Several brand names from one kitchen, separate menus, shared stock	Menu	Phase 3
Gift cards and vouchers	Sell and redeem, balance tracked	Payments	Phase 3
Corporate accounts	Offices order on credit, one monthly bill	Customers	Phase 3
Self-order kiosk	Touch screen ordering and payment	Selling	Phase 3
Waitlist	Queue for busy restaurants with WhatsApp alert when table is free	Dine-in	Phase 3
Accounting export	QuickBooks and Xero for international customers	Finance	Phase 3
Open API and webhooks	Other apps connect to Servio	Platform	Phase 3
Franchise mode	Head office controls menu and prices; franchises report sales	Multi-branch	Phase 3

Feature	What it does	Module	Phase
Reseller program	Agents in other cities sell Servio for commission	Platform	Phase 3
Menu change approval	Manager suggests, owner approves (chains)	Menu	Phase 3
Allergen and dietary tags	Shown on menu and receipts (needed in UK and Gulf)	Menu	Core
Opening and closing checklists	Staff tick tasks at start and end of day	Staff	Phase 2
In-app setup tour and notification center	Guides new users; one place for all alerts	Platform	Core

Core business logic

These rules make the numbers trustworthy. Every module follows them.

1. Order life cycle

Draft → Placed (KOT sent) → Preparing → Ready → Served or Dispatched → Paid → Closed. Branches: Held (parked), Voided (before payment, needs reason and PIN), Refunded (after payment, needs manager).

- Stock is deducted once, at **Placed**, so the kitchen and the stock figure match.
- A void before cooking returns the stock. A void after cooking records it as waste.
- Revenue counts only at **Paid**. Open orders show separately ("Rs 8,340 still open").
- Each order gets a short daily serial (#59) for people and a unique ID for the system.

2. Stock deduction

- Each menu item has a recipe. Each sold quantity deducts recipe quantity x order quantity, converted into the stock unit.
- Sub-recipes (sauce, dough) are produced in batches through a Production action, then deducted like normal items.
- Packaging is added by order type (delivery box, takeaway bag).
- Stock can go below zero from sales, but this creates an alert and a task for the manager.

- Every change writes a ledger row: item, section, before, after, cost, document, user. Ledger rows are never edited, only reversed.

3. Cost

- Item cost uses weighted average cost, updated on every goods received.
- Food cost % = recipe cost / selling price (without tax). Shown on each menu item; red above a target (e.g. 35%).
- Labor, gas and rent are overheads, not stock.

4. Cash drawer and shifts

1. Cashier opens a shift with opening cash.
2. Every cash payment, refund, cash in and cash out (with reason) updates expected cash.
3. At close, the cashier counts cash without seeing the expected amount (blind count).
4. The system shows over or short; the manager approves.
5. Z report prints: sales by payment type, voids, discounts, tax, cash over/short.

5. Tax

- Tax rate set per item or category, with inclusive or exclusive pricing.
- On payment, the invoice is sent to FBR or PRA. The returned invoice number and QR print on the receipt.
- If offline, invoices queue and sync later; the queue is visible on the dashboard.

6. Day close

All shifts closed → pending orders resolved → stock count (optional) → day locked. After locking, only a manager with reason can reopen.

Accounts, login and data isolation

Every restaurant sees only its own data. This is enforced by the server and the database, never by the URL or the screen.

Account structure

- **Business (tenant):** the paying customer, e.g. "Karachi Grill". Holds plan, billing, menus, staff, customers.
- **Branch:** a location of that business. Holds tills, tables, stock sections, tax registration, opening hours.

- **User:** one person with one login (phone or email). A user can belong to several businesses.
- **Membership:** links a user to a business with a role (Owner, Manager, Cashier, Waiter, Kitchen, Rider) and the branches they can access.
- **Device:** a till, tablet or kitchen screen registered to one branch. Staff log in on it with a 4-digit PIN.

Domain decision: one main app domain for staff, subdomains for customers

Who	URL	Why
Marketing site	servio.com, servio.pk	Public website, pricing, signup
Owners, managers, staff	app.servio.com	One login for everyone; a user in two businesses picks one from a switcher. Simpler to build, secure cookies on one domain
Till and kitchen devices	Servio POS desktop app (Windows, Mac) and Android tablet app, signing in to app.servio.com	Device is paired to a branch with a one-time code
Customers: online ordering and QR menu	karachigrill.servio.menu (slug subdomain)	Each restaurant gets its own branded ordering page; easy to share and print on QR codes
Customers on Business plan	order.karachigrill.com (custom domain)	Restaurant's own domain, connected by a DNS record; SSL issued automatically
Super admin	admin.servio.com	Separate login with 2FA, office IP allowlist

We do not put the staff panel on per-restaurant subdomains. It adds wildcard login cookies, harder local development and confusing logins for users in several businesses, with no security gain, because isolation comes from the server (below).

Isolation rules (must follow)

1. Every business-owned table has a `business_id` column, not null, indexed.
2. After login, the server sets the current business from the user's membership, never from a URL parameter or form field.
3. All queries are scoped automatically (e.g. the acts_as_tenant gem). Unscoped queries are blocked outside super admin code.
4. PostgreSQL Row-Level Security is the second wall: even a coding mistake cannot return another business's rows.
5. Branch-level access is checked too: a cashier of Branch A cannot see Branch B.
6. IDs in URLs are UUIDs, not 1, 2, 3, so they cannot be guessed.
7. Files (photos, receipts, exports) are stored under the business folder and served by short-lived signed links.
8. An automated test suite tries to read and write another business's data on every endpoint, and must fail every time.
9. Super admin can open a business only for support, with the owner's approval, and every action is logged.

Login methods

- Owners and managers: phone + WhatsApp OTP, or email + password; 2FA optional (required for owners on Business plan).
- Staff on devices: 4-digit PIN; device must be registered to the branch.
- Customers (online ordering): phone + WhatsApp OTP, no password.
- Sessions expire after inactivity; owner can log out any device remotely.

WhatsApp integration

Servio uses the official WhatsApp Business Platform (Meta Cloud API), never unofficial WhatsApp Web tools, which get numbers banned.

Use cases

Message	To	Trigger	Type	Phase
Login OTP	Owner, staff, customer	Login or signup	Authentication	Core

Message	To	Trigger	Type	Phase
Digital receipt with PDF link	Customer	Payment, if customer gave phone	Utility	Core
Order status: confirmed, ready, out for delivery	Customer	Order status change	Utility	Core
Daily sales summary	Owner	Day close, or 11 pm	Utility	Core
Alerts: big void, cash short, low stock	Owner, manager	Event over a set limit	Utility	Core
Trial and billing reminders	Owner	Day 60, 80, 88; invoice due	Utility	Core
Reservation confirmation and reminder	Customer	Booking, 2 hours before	Utility	Phase 3
Feedback request	Customer	1 hour after order	Marketing	Phase 3
Campaigns: offers, birthday deals	Opted-in customers	Owner sends from Customers page	Marketing	Phase 3
Order by WhatsApp chat (AI bot)	Customer	Customer messages the restaurant	Service	Phase 4

Two ways to send

1. **Shared Servio number** (Starter and Pro): messages come from Servio's verified number with the restaurant name in the text. Zero setup for the restaurant.
2. **Restaurant's own number** (Business plan): the owner connects their WhatsApp Business number through Meta Embedded Signup inside Servio settings. Needs Servio approved as a Meta Tech Provider.

Rules for developers

- All business-initiated messages use pre-approved templates, in English and Urdu (Arabic later). Keep a template list in the database with Meta approval status.

- Marketing messages only to customers with recorded opt-in (date, source); every message has an opt-out reply ("STOP"), which updates the customer record.
- Sending happens in background jobs with retries; delivery and read statuses come back by webhook and are stored per message.
- Each business has a monthly message counter; utility messages over the plan limit and all marketing messages use prepaid credits.
- Meta charges per delivered template message, by country and category. Check current rates before launch and keep prices in a settings table.
- Fallback: if WhatsApp fails, OTP goes by SMS and receipts by email.

Data model

About 70 core tables in 9 groups. All except the platform group carry `business_id`; branch-level ones also carry `branch_id`.

Group	Tables
Platform	businesses, branches, users, memberships, devices, plans, subscriptions, invoices, payments_received, feature_flags
Menu	menus, categories, products, product_variants, modifier_groups, modifiers, deals, deal_items, promotions, price_overrides (per branch)
Dine-in	floors, tables, reservations
Orders	orders, order_items, order_item_modifiers, order_status_events, payments, refunds, voids, discounts_applied
Money	shifts, cash_movements, tax_rates, tax_invoices (FBR/PRA), expenses, expense_categories, accounts
Stock	stock_items, units, unit_conversions, stock_sections, stock_levels, stock_ledger_entries, stock_documents, recipes, recipe_lines, production_batches

Group	Tables
Purchasing	suppliers, purchase_orders, purchase_order_lines, goods_receipts
People	customers, customer_addresses, loyalty_transactions, riders, deliveries, employees, attendance, payroll_runs, payslips, advances
System	audit_logs, whatsapp_messages, whatsapp_templates, notifications, import_jobs, sync_batches

Rules every developer must follow

1. **Money** is stored as integers in the smallest unit (paisa, cents), with the business currency. Never floats.
2. **Quantities** are stored as decimals with 3 places in the item's base unit (g, ml, pcs).
3. **Time** is stored in UTC; shown in the branch time zone.
4. **Business day:** each branch has a day-end time (e.g. 4:00 am). An order at 1:30 am belongs to the previous business day in every report.
5. **Order numbers:** a short daily serial per branch (#59) for people, plus a UUID for the system. Tax invoice numbers come from FBR/PRA.
6. **Nothing financial is deleted.** Orders, payments, ledger rows and shifts are only voided or reversed. Menu items and staff are archived, not deleted.
7. **Price snapshots:** order items save name, price, tax and recipe cost at the time of sale, so later menu changes never change old reports.
8. **Offline IDs:** records created on a till get a UUID on the device; the server accepts each UUID once (idempotent sync).
9. **Audit log** row for every create, update, void, refund, discount, price change, permission change and login, with user, device and before/after values.
10. **Soft limits from the plan** (branches, devices, staff, messages) are checked on the server.

Roles and permissions

Seven default roles (the six below plus Accountant, who sees reports, expenses and finance only), each editable. Staff log in with a 4-digit PIN on the till; managers approve sensitive actions with their own PIN.

Action	Owner	Manager	Cashier	Waiter	Kitchen	Rider
Take orders	Yes	Yes	Yes	Yes	No	No
Take payment	Yes	Yes	Yes	No	No	Cash on delivery only
Discount up to 10%	Yes	Yes	Yes	No	No	No
Discount above 10%	Yes	Yes	Manager PIN	No	No	No
Void order	Yes	Yes	Manager PIN	No	No	No
Refund	Yes	Yes	No	No	No	No
Open and close shift	Yes	Yes	Own shift	No	No	No
Kitchen display	View	Yes	View	View	Yes	No
Edit menu and prices	Yes	Optional	No	No	No	No
Stock actions	Yes	Yes	No	No	Waste only	No
Reports	All	Branch	Own shift	No	No	Own deliveries
Payroll and settings	Yes	No	No	No	No	No

All permission checks run on the server, not only in the screen.

Three apps, one login system

Staff use three apps. One account system serves all three; the role decides which app a person can open and what they can do in it.

App	Device	Who can log in	Login method
Admin panel	Web browser on laptop or phone	Owner, Manager, Accountant	Phone + WhatsApp OTP, or email + password
POS app (cashier and order taker)	Desktop or tablet at the counter; waiter's tablet or phone	Cashier, Waiter, Manager, Owner	4-digit PIN on a registered device
Rider app	Rider's mobile phone (Android, iOS)	Rider	Phone + WhatsApp OTP

A person opening an app their role does not allow sees "You don't have access to this app" and is not logged in.

How the owner gives permission

1. Admin panel → Staff → Add staff.
2. Enter name and phone number.
3. Choose a role (Manager, Cashier, Waiter, Rider, Kitchen, Accountant, or a custom role).
4. Choose the branch or branches the person works at.
5. Servio sends a WhatsApp invite. Cashiers and waiters get a 4-digit PIN for the POS app; riders get the Rider app link; managers get an admin panel login.
6. Deactivate removes access at once on every app (offline devices on their next sync).

Custom roles

The owner can copy a role and tick or untick permissions, for example a "Senior Cashier" allowed 20% discounts. Owner permissions cannot be removed from the last owner.

Security rules

- **Manager approval on the spot:** a manager types their own PIN on the cashier's screen to approve a void, refund, big discount or price override. Both names are saved.
- **Registered devices:** the POS app works only on devices paired to the branch by the owner or manager with a one-time code.
- **Server checks every action:** the screen hides buttons a role cannot use, but the server checks again before saving.

Technical design

- Tables: users, roles, permissions, role_permissions, memberships (user + business + role + branches), devices.
- Each permission has a code, e.g. `app.admin`, `app.pos`, `app.rider`, `orders.create`, `orders.void`, `orders.refund`, `discounts.above_limit`, `menu.edit`, `staff.manage`, `reports.view_all`, `reports.view_own`.
- Every API request checks: user active, member of this business, allowed at this branch, has this permission, using an allowed app.
- Every sensitive action writes an audit log row with who did it and who approved it.

UI and UX guidelines

The till is built for speed and touch; the admin panel is built for clarity. Both share one design system.

Till screen (cashier)

- Three columns: categories (left), item grid with photos (middle), current order and pay button (right).
- Touch targets at least 48 px. Works on 10-inch tablets and 1366 px laptops.
- Big, colored order-type switch at the top: Dine-in, Takeaway, Delivery.
- One-tap repeat of a customer's last order by phone number.
- Dark mode for night shifts; high contrast for bright counters.
- Keyboard shortcuts for laptop cashiers (F1 new order, F2 search, F8 pay).

Admin panel (owner, manager)

- Short sidebar: 7 groups (Sell, Menu, Money, Stock, People, Reports, Settings). Modules not switched on are hidden.
- Every page starts with 3 to 4 summary cards, then filters, then the list.
- Numbers: PKR with commas (Rs 1,487,366), max 3 decimals, units always shown.
- Empty states explain the next step and offer a sample-data button.
- Charts: one clear title, correct axis units, today vs same day last week.

Visual style

- One primary color (deep indigo or teal), one accent, green for money in, red for money out and alerts.
- Font: Inter for English, Noto Nastaliq Urdu for Urdu receipts and labels.

- Real food photos on items and deals; a default illustrated placeholder per category, never an empty box.
- Consistent page header: icon, title, one-line description, actions on the right.

Receipts

- 58 mm and 80 mm thermal printers, English or Urdu.
- Logo, serial number, items, tax, FBR/PRA QR, payment method, WhatsApp number for feedback.

Localization and hardware

Servio is built once and configured per country, so the same code serves Lahore, Dubai and London.

Localization

- Languages: English and Urdu at launch (Urdu right-to-left layout tested), Arabic later. All screen text in translation files, none hard-coded.
- Currency per business; formats per locale (Rs 1,487,366 / AED 1,250.00 / £12.50).
- Tax engine per country: Pakistan FBR and provincial (PRA, SRB, KPRA, BRA), UAE VAT, UK VAT. Rate, inclusive or exclusive, and invoice rules are settings, not code.
- Payment methods per country: Pakistan cash, card, EasyPaisa, JazzCash, bank; elsewhere cash and card terminals.
- Time zone and business day-end per branch.
- Phone numbers stored in international format (+92…).

Hardware

Device	Support
Till	Servio POS desktop app on Windows or Mac laptop/PC; Android tablet (10 inch+) app; iPad later
Kitchen display	Any Android TV box or tablet in browser kiosk mode
Receipt printers	58 mm and 80 mm ESC/POS thermal, USB or network (LAN/Wi-Fi); Urdu printed as image

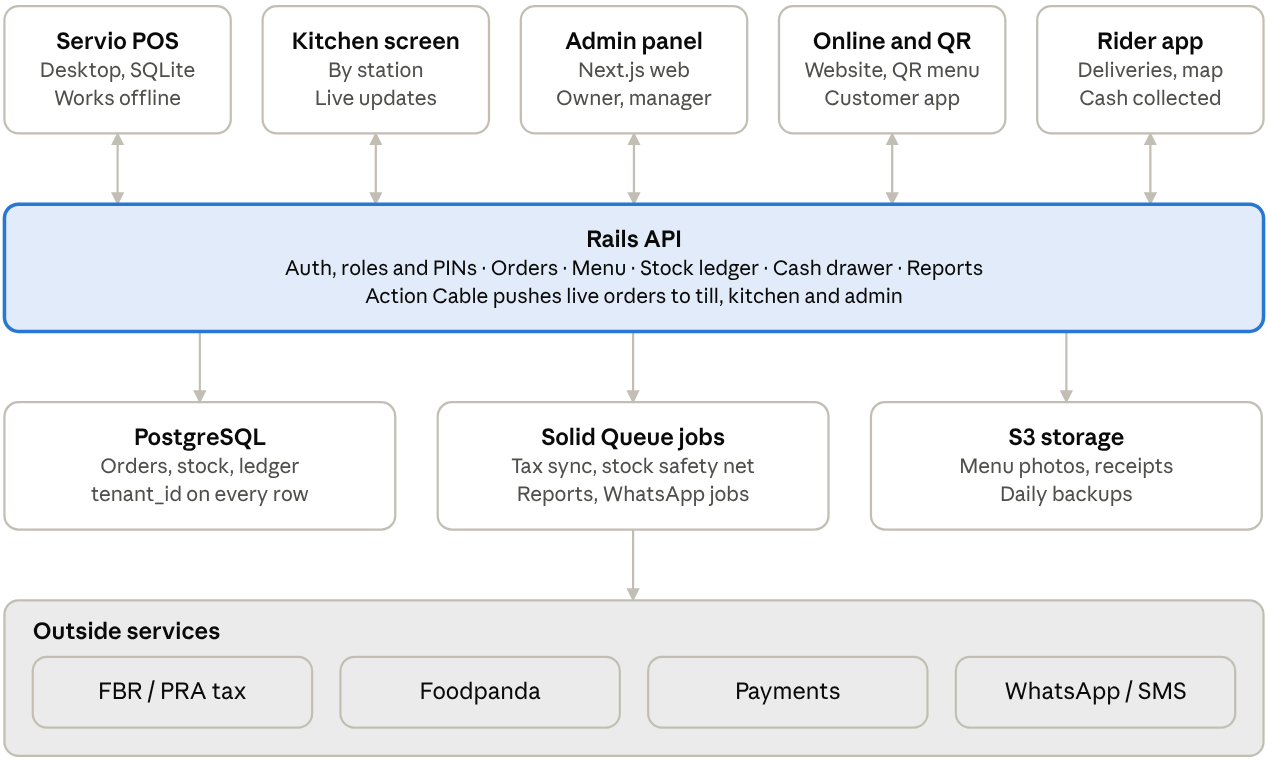
Device	Support
Kitchen printers	One per station, network; prints KOT when kitchen display is off
Cash drawer	Opens through the receipt printer port
Barcode scanner	USB keyboard-mode, for retail items and stock
Customer display	Second screen showing order and total (Phase 2)
Fingerprint device	Common ZKTeco models, for attendance (Phase 3)

The desktop till prints directly to USB and network printers. A small print bridge app does the same for Android tablet tills.

Tech stack and architecture

One Rails API serves the till, kitchen screen, admin panel, online ordering and rider app. These are proven, well-known tools that are easy to hire for.

One Rails API serves every screen; background jobs talk to outside services



system architecture · 5 apps, 1 API, 3 data stores, 4 outside services

The till saves every order locally first with its own ID, then syncs. If the same order is sent twice, the API keeps one copy. Full details in Apps and offline sync below.

Layer	Choice	Why
Backend API	Ruby on Rails 8, API mode	Fast to build; mature auth, jobs, WebSockets
Admin panel	Next.js, Tailwind, shadcn/ui	Modern, attractive, fast
Till and kitchen	Desktop app (Tauri) with local SQLite, using the same Next.js screens; kitchen display in the browser	Fast, prints directly, keeps selling through load-shedding and internet cuts
Mobile apps	React Native (Expo)	Rider app first; branded customer app later
Database	PostgreSQL 16, business_id on every table, Row-Level Security	One database for all restaurants

Layer	Choice	Why
Background jobs	Solid Queue (built into Rails 8); Sidekiq and Redis later if load grows	Tax sync retries, stock safety net, scheduled reports
Real-time	Action Cable	New orders reach the kitchen instantly
Printing	ESC/POS; desktop till prints directly, small print bridge for Android tablets	58 mm and 80 mm thermal printers
Hosting	AWS Bahrain region (me-south-1) with Kamal 2, or Heroku for the pilot; S3, daily backups	Closest region to Pakistan and the Gulf; cheap start, easy scale
Monitoring	Sentry	Catch till and API errors early

Apps and offline sync

Servio ships as a desktop till, two mobile apps and web pages, all on one API. The till keeps selling when the internet is down and syncs when it returns.

Apps

App	Platform	Who uses it	Main features	Offline	Phase
Servio POS (till)	Windows and Mac desktop (Tauri), Android tablet, waiter's phone	Cashier, manager	Orders, payments, printing, cash drawer, shifts, voids	Full	Core
Kitchen display	Browser on Android TV box or tablet	Kitchen staff	Orders by station, timers, bump	Through local hub (Phase 3)	Core
Admin panel	Web (app.servio.com)	Owner, manager	Menu, staff, stock, reports, settings, billing	No	Core
Rider app	Android and iOS (React Native, Expo)	Rider	Assigned deliveries, map and directions, status updates, cash collected, end-of-shift settlement	Queue and sync	Phase 2

App	Platform	Who uses it	Main features	Offline	Phase
Customer ordering	Web page per restaurant (subdomain), works like an app	Customers	Menu, cart, pay, track order, loyalty points	No	Phase 2
Branded customer app	Android and iOS, restaurant's own name and logo	Customers	Same as customer ordering, plus push offers	No	Phase 3, Business plan add-on

Waiters take orders with the POS app on a tablet or phone. Owners use the admin panel, which works on phones, plus WhatsApp alerts; a separate owner app can come later.

How offline works on the till

1. The till keeps a local SQLite database: menu, prices, tax, staff PINs, tables, open orders and the current shift.
2. Every action (order, payment, void, cash in/out) is saved locally first, in under 100 ms, whether online or not.
3. The app checks the connection every few seconds and shows a badge: Online (green), Offline (orange), Syncing (blue).
4. A sync queue sends saved records to the API in the order they happened. Each has a UUID made on the device; the API accepts each UUID once, so nothing is duplicated.
5. Changes made in the admin panel (prices, new items, staff) come down to the till on the next sync.
6. If a record fails on the server (e.g. item was deleted), it goes to a "needs attention" list for the manager, never lost silently.

Offline rules

Works offline	Needs internet (queued or blocked)
Taking orders, cash payments	EasyPaisa, JazzCash, online card payments (blocked)
Printing receipts and kitchen tickets	Online, QR and Foodpanda orders (cannot arrive)
Cash drawer, shift open and close	WhatsApp messages (queued)
Discounts and voids with manager PIN	FBR/PRA invoice number (queued; receipt reprinted with QR after sync, or shown on the digital receipt)

Works offline	Needs internet (queued or blocked)
Today's report for that till	Live owner dashboard

Rules for developers

- **Order numbers offline:** each till has a code (T1, T2). Offline serials are T1-059, T2-059, so two tills never clash.
- **Stock** is calculated on the server from synced sales; the till only shows the last known stock.
- **Price conflicts:** an offline order keeps the price the till had at the time of sale.
- **Staff PIN change** made online reaches tills on the next sync; a removed staff member is blocked on sync.
- **Data safety:** the local database is encrypted; synced orders older than 7 days are cleared from the device.
- **Updates:** the desktop app updates itself automatically, but never during an open shift.
- **Local hub (Phase 3):** one till acts as a hub on the shop Wi-Fi, so other tills, waiter tablets and the kitchen display keep talking to each other when the internet is down.

Security, backups and performance

Restaurants trust Servio with their cash and customer data, so these are requirements, not nice-to-haves.

Security

- HTTPS everywhere; passwords hashed (bcrypt); PINs hashed and locked after 5 wrong tries.
- Rate limits on login, OTP and public ordering endpoints (e.g. Rack::Attack).
- Role checks on every API endpoint (e.g. Pundit), plus the tenant isolation rules above.
- Secrets in AWS Secrets Manager or Rails credentials, never in the repo.
- Payment card data never touches Servio servers (Paddle and card terminals handle it).
- Dependency and security scans in CI (Brakeman, bundler-audit, npm audit).
- Yearly external penetration test once there are paying international customers.

Backups and data ownership

- Automatic daily database backups kept 30 days, plus point-in-time recovery for 7 days; restore tested monthly.

- Uploaded files versioned in S3.
- Owners can export all their data (menu, orders, customers, stock) as CSV or Excel at any time.
- On account deletion, data is removed within 30 days, except invoices kept as tax law requires.
- Privacy: customer data used only by that restaurant; follow Pakistan's data protection rules and GDPR for UK/EU customers.

Performance and reliability targets

Target	Value
Add item on till	under 100 ms (local, works offline)
Payment and receipt print	under 2 seconds
Order reaching kitchen display	under 2 seconds online
Admin page load	under 2 seconds on 4G
Offline operation	at least 24 hours of orders stored on the till
Uptime	99.9% per month, shown on a public status page
Scale for Year 1	1,000 businesses, 2 million orders per month

Development process

Separate repositories per app, small pull requests, and nothing reaches customers without passing tests on staging.

Repositories

- `servio-api` : Rails API, background jobs, super admin.
- `servio-web` : Next.js admin panel, kitchen display and online ordering pages, plus the shared till screens (one monorepo with shared UI components is fine).
- `servio-pos-desktop` : Tauri desktop till with local SQLite and sync queue.
- `servio-mobile` : React Native Rider app and branded customer app (Phase 2+).
- `servio-print-bridge` : printing helper for Android tablets and kitchen printers (the desktop till prints directly).

Environments

Environment	Purpose	Data
Local	Developer machines (Docker Compose)	Seed data
Staging	Testing and client demos	Fake businesses, copy of real menu structure
Production	Live customers	Real data, backups on

Testing

- Unit and request tests for every API endpoint (RSpec); the tenant isolation test runs on all of them.
- Money and stock logic covered by tests with fixed examples (an order, a void, a refund, a shift close).
- End-to-end tests for the main flows: signup, take order, pay, print, void, close shift (Playwright).
- Offline test: disconnect the till, take 20 orders, reconnect, confirm no duplicates and correct stock.

CI/CD

- GitHub Actions: lint, tests, security scan on every pull request.
- Merge to main deploys to staging automatically; production deploy by tag, during low-traffic hours (3 pm to 5 pm Pakistan time).
- Database migrations must be backward compatible; feature flags for unfinished features.
- Errors to Sentry; logs and uptime alerts to the team's WhatsApp or Slack.

Definition of done (for every feature)

- [ ] Works on 1366 px laptop, 10-inch tablet and phone
- [ ] English and Urdu text in translation files
- [ ] Permissions checked on the server
- [ ] Tenant isolation test passes
- [ ] Audit log written for money, stock and permission changes
- [ ] Tests written and passing in CI

- [ ] Tested on staging by someone other than the developer
- [ ] Help text or help article updated

API rules

All apps talk to one REST API that follows the same rules, so every developer builds endpoints the same way.

- **Versioning:** all endpoints under `/api/v1/`. Breaking changes go to `/api/v2/`; old versions stay live until every till has updated.
- **Format:** JSON, snake_case fields, money as integers in the smallest unit with a `currency` field, times in ISO 8601 UTC.
- **Errors:** one shape everywhere: `{"error": {"code": "orders.already_paid", "message": "...", "details": {...}}}` with the right HTTP status (400, 401, 403, 404, 409, 422, 429).
- **Pagination:** cursor based (`?cursor=...&limit=50`, max 200).
- **Idempotency:** every create request from tills and apps sends an `Idempotency-Key` (the record's UUID); repeats return the first result.
- **Auth:** short-lived access token plus refresh token; device tokens for POS devices; tokens carry user, business, branch and app.
- **Documentation:** OpenAPI spec generated from tests (rswag); TypeScript types generated from it for the web and apps.
- **Webhooks (Phase 3):** signed with a secret, retried with backoff, for order created, order paid, stock low.
- **Rate limits:** per user and per device; public ordering endpoints stricter.

Launch checklist

These must be ready before the first paying customer, beside the Phase 1 features.

Moving customers from ArfaPOS and other systems

- [ ] Import menu (categories, products, prices, images) from Excel/CSV, using ArfaPOS Catalog export
- [ ] Import stock items with units and opening quantities, using ArfaPOS Items export
- [ ] Import customers (name, phone, address)
- [ ] Menu import from a photo of the printed menu card (AI, Phase 4; manual service until then)

- [ ] Import preview with error list before anything is saved

Legal

- [ ] Company registration (SECP) and bank account for receiving payments
- [ ] Terms of service, privacy policy, refund policy, data processing terms for UK/EU
- [ ] Trademark application for "Servio"
- [ ] FBR/PRA integration approval for Servio as a POS provider
- [ ] Meta Business verification for WhatsApp

Support

- [ ] Help center with short videos in English and Urdu
- [ ] In-app chat and a support WhatsApp number, with support hours
- [ ] Public status page
- [ ] Onboarding call or visit for the first 20 customers

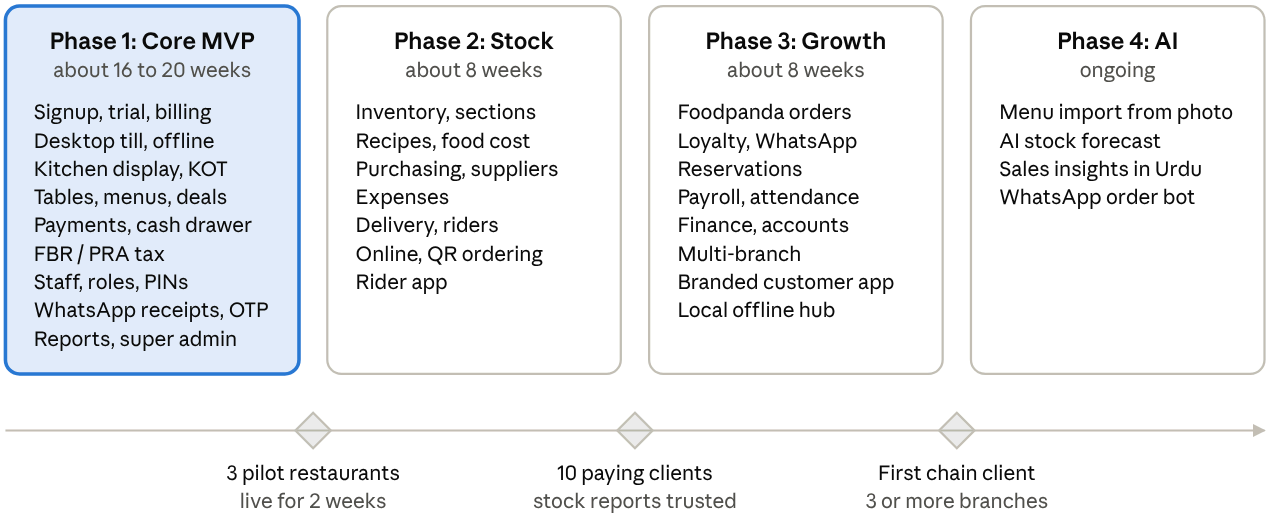
Marketing site (servio.com / servio.pk)

- [ ] Home, features, pricing, signup, demo booking, contact
- [ ] Pakistan page with PKR prices and Urdu tagline
- [ ] Demo account with sample restaurant data
- [ ] Case study from the first pilot restaurant
- [ ] Referral program: 1 free month for each restaurant referred

Build roadmap

Ship a sellable core POS, including signup, billing and WhatsApp basics, in about 16 to 20 weeks, then add one module group at a time. Durations are estimates for a team of 2 to 3 developers.

Launch the core till first; stock, growth and AI follow in phases



build roadmap · 4 phases, 3 gates

Each diamond is a gate: the next phase starts only when the one before is proven with real restaurants.

Open questions

- First target segment for pilots: fast food, desi dine-in, or cafes
- Pilot restaurants: which 3 (own restaurant first?)
- Hardware bundle: sell printers and tablets, or software only
- Team: number of developers and monthly budget
- First international market: UAE, UK, or Saudi Arabia